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Input Impedance and Stability Analysis of VSC in Integrated On-Board Chargers for Electric Vehicles

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***Abstract*—This paper deals with a three-phase integrated on-board charger of electric vehicles (EVs) that reutilize the traction motor's stator winding as part of the input filter to form an LCL filter for the front-end voltage source converter (VSC) in dual-stage chargers. This approach imposes additional constraints on the selection of one filter inductance, based on motor specifications, which may affect the control loop stability dynamics, particularly under grid impedance variations. To examine this, the small signal model of the LCL-filtered VSC is developed and its d-q frame based impedance model is derived. The control loops, including the active damping, current, and voltage, are designed, and overall input impedance characteristics are analysed and compared with typical LCL filter based VSC. The system's stability is further assessed through generalized Nyquist criterion under grid impedance variations, and analytical findings are validated in real-time Hardware-in-the-Loop testing.**

***Index Terms*—Electric vehicles, Impedance model, Integrated on-board charger, LCL filter, Stability, Voltage source converter.**

I. Introduction

With the surge in EV demand, the major challenge for its adoption is the charging infrastructure, primarily due to costs and availability. On-board chargers (OBCs), an economical alternative to high-cost off-board DC chargers, integrate charging components within the vehicle, allowing for convenient charging from AC outlets. OBCs can be single or dual-stage, with the latter often preferred for its broader output voltage range, featuring a front-end boost rectifier for unity power factor, and a DC-DC buck converter to match the battery voltage level. However, their main limitations include slow charging rates and the bulky size of faster-charging components, which pose a challenge due to onboard space limitations. Research into compact, faster-charging solutions is essential for broader EV adoption [1].

Integrated OBCs can efficiently use on-board space by repurposing EV traction components, such as inverters and motors, in the charging circuit, thereby gaining research interest. The reutilization of the EV motor's stator winding as part of the input filter, with the traction inverter acting as a front-end VSC through a reconfiguration circuit during charging [2], [3], forms an LCL filter that offers superior switching harmonics rejection over an L-filtered VSC [4]. However, the motor winding's equivalent inductance during charging differs from that of an independently designed typical LCL filter, leading to variations in the input impedance characteristics of the converter as perceived by the power grid. This may alter control performance and control loop stability. Additionally, variations in grid impedance could provoke distinct stability responses compared to those with a typical LCL filter when a feedforward loop is employed. It is, therefore, crucial to examine the impedance characteristics of an integrated LCL-filtered VSC and contrast them with a typical LCL filter design to identify appropriate compensation strategies, ensuring an effective VSC design for IOBCs.

In the literature, impedance methods for three-phase balanced systems are primarily divided into sequence-based and dq frame-based methods [5], with the latter frequently used by several authors for stability analysis of GCC systems [6]–[9]. In [6], the dq impedance model of GCC systems is developed to analyze and specify input impedance for stability at AC terminals under high power factor load assumptions. The input

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impedance of the converter system is investigated under constant power load conditions in [7] through small-signal impedance-based stability analysis. The impact of the DC voltage control loop using the dq impedance model has revealed low-frequency and high-frequency oscillations in rectifier and inverter modes, respectively. The impact of decoupled control along with the current loop and phase-locked loop (PLL) is considered in [8] for dq impedance analysis of the inverter and has shown enhanced stability characteristics. The author in [9] developed an equivalent single-input single-output dq impedance model from a multi-input multi-output impedance model, considering the PLL and decoupling terms. This made the analysis simpler and more intuitive by applying the classic Nyquist criterion instead of the generalized Nyquist criterion. However, it is important to note that the discussed literature deals with typical L or LCL filter designed independently to fulfill desired control performance objectives. Therefore, it is imperative to investigate the impedance stability characteristics of the converter system when the integrated LCL filter design is considered, as in the case of IOBC. This article aims to fulfill this research gap.

This paper begins with a system description of considered three phase IOBCs in Section II, followed by small-signal modeling and control design in Section III. Input impedance model derivation and stability analysis with various control loops are presented in Section IV. HIL verification results are provided in Section V, and the conclusion is given in Section VI.

II. Three Phase IOBC System

Fig. 1 shows schematic diagram of a dual-stage IOBC. The power stage includes a utility grid , an integrated LCL input filter, a three-phase VSC for power factor correction (PFC), and a DC/DC converter connected to an EV battery. For simplicity and due to its minimal impact on the first-stage converter and its control design, the DC/DC converter and battery are modeled as a load resistance R_{dc}. The DC-link capacitor C_{dc} is used to suppress the ripples in output DC voltages. The LCL filter include inductances L_1 and L_2, resistance R_1 and R_2 and filter capacitance C_f. Contrary to typical LCL filter designs, where each component is designed separately to optimize input current quality and power factor, this system incorporates the reconfigured stator winding of EV traction motor (i.e. induction motor (IM) in this case) as grid-side filter component, thus named motor integrated LCL filter. However, the value of inductance L_2 and its equivalent resistance R_2 depends on IM specifications. The detail regarding stator winding reconfiguration during charging mode and reconfiguration circuit to switch between charging and traction mode can be found in [10].

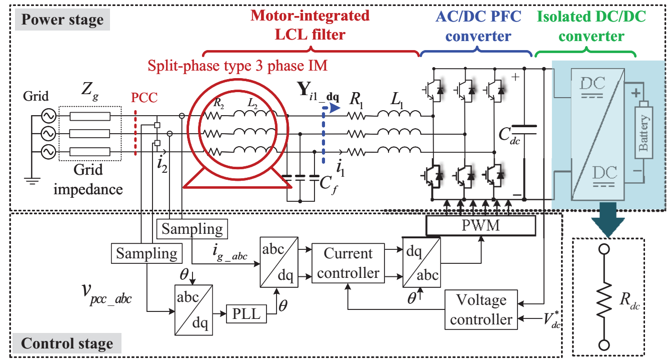


Fig. 1. Circuit diagram of 3ph LCL-filtered VSC IOBC system

The control stage regulates output DC voltage and grid current by measures DC link voltage V_{dc}, grid currents i_{g_abc}, and voltage at point of common coupling v_{g_abc}, where Z_g denotes grid impedance. It includes voltage and current loops and a PLL block for synchronization, employing Park transformations for dq components conversion, proportional integral (PI) controllers to minimize errors, and pulse width modulation (PWM) for converter switching signals. In charging mode, without electromagnetic force, the IM's winding acts as a filter with inductance L_2 and resistance R_2, as shown in Fig. 1, crucial for controller design. These values are obtained through experimental set-up using a special designed split phase type IM with a three-phase resistive load as shown in Fig. 2, reconfigured into an open-end winding. Measurements of voltage, current, power factor, and resistive load voltage per phase are taken with a power analyzer to calculate the stator winding's equivalent inductance and resistance in charging mode. An IM with 11 kW power rating, speed of 1470 R/min, power factor of 0.85, and a torque of 71.5 Nm, have been utilized and the calculated equivalent values of L_2 and R_2, are detailed in Table I.

III. Modelling and Control Structure Design

The small signal model of a three-phase VSC with an integrated LCL filter is obtained by combining the L-filtered boost rectifier model with the L_2C_f filter component. The model remains consistent for both integrated and typical LCL filters due to the uniform design of the PFC converter. The boost rectifier's small signal model is obtained by linearizing its average model at the operating point and applying a three-phase to dq transformation

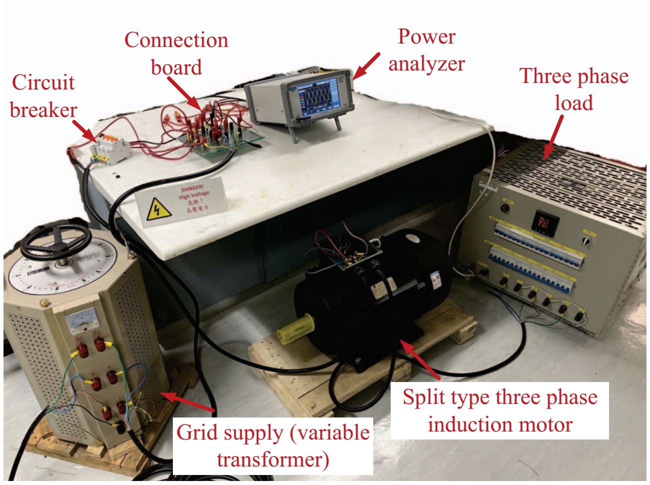


Fig. 2. Experimental set-up for L_2 and R_2 measurement of IM

for voltage and current, facilitating control and stability analysis. The transfer functions (TFs) and their matrices are then derived, with perturbed signals denoted by lowercase letters with symbol (ˆ) and operating point steady-state values by uppercase letters. The plant's TF matrices and their detailed expressions relating input voltages ($\hat{v}_d$ and $\hat{v}_q$) to inverter-side current ($\hat{i}_{1d}$ and $\hat{i}_{1q}$), and control signals ($\hat{d}_d$ and $\hat{d}_q$) to inverter-side current, are given as follows

$$\begin{bmatrix} \hat{i}_{1d} \\ \hat{i}_{1q} \end{bmatrix} = \overbrace{\begin{bmatrix} Y_{i_1\_dd} & Y_{i_1\_dq} \\ Y_{i_1\_qd} & Y_{i_1\_qq} \end{bmatrix}}^{\mathbf{Y}_{i_1\_\mathbf{dq}}} \begin{bmatrix} \hat{v}_d \\ \hat{v}_q \end{bmatrix} \tag{1}$$

$$\begin{bmatrix} \hat{i}_{1d} \\ \hat{i}_{1q} \end{bmatrix} = \overbrace{\begin{bmatrix} G_{i_1d\_dd} & G_{i_1d\_dq} \\ G_{i_1d\_qd} & G_{i_1d\_qq} \end{bmatrix}}^{\mathbf{G}_{i_1d\_\mathbf{dq}}} \begin{bmatrix} \hat{d}_d \\ \hat{d}_q \end{bmatrix} \tag{2}$$

$$Y_{i_1\_dd} = \frac{3Z_{dc}D_q^2 + 2Z_{L_1}}{3Z_{dc}Z_{L_1}D_d^2 + 3Z_{dc}Z_{L_1}D_q^2 + 2\omega^2L_1^2 + 2Z_{L_1}^2};$$
$$Y_{i_1\_qq} = \frac{3Z_{dc}D_d^2 + 2Z_{L_1}}{3Z_{dc}Z_{L_1}D_d^2 + 3Z_{dc}Z_{L_1}D_q^2 + 2\omega^2L_1^2 + 2Z_{L_1}^2}.$$

where,

$$Z_{L_1} = L_1s + R_1; \quad Z_{dc} = \frac{R_{dc}}{R_{dc}C_{dc}s + 1}.$$

In the matrices above, only diagonal TFs are considered due to their dominant role on the open-loop gain compared to off-diagonal elements, especially when high power factor load operation is assumed. Further, the TFs between control to output DC voltage are calculated as

$$\hat{v}_{dc} = \overbrace{\begin{bmatrix} G_{vd\_d} & G_{vd\_q} \end{bmatrix}}^{\mathbf{G}_{vd\_\mathbf{dq}}} \begin{bmatrix} \hat{d}_d \\ \hat{d}_q \end{bmatrix}; \tag{3}$$

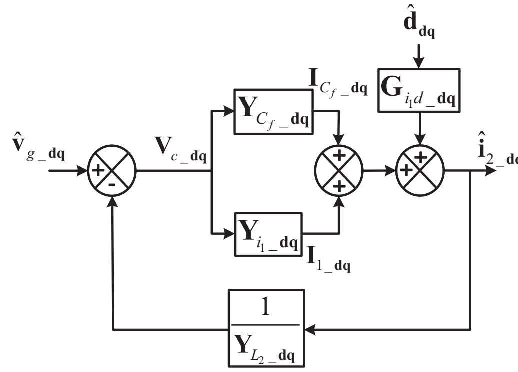


Fig. 3. Equivalent block diagram of LCL-filtered VSC.

$$G_{vd\_d} = -\frac{3Z_{dc}(D_dV_{dc} - I_dZ_{L1})}{3D_d^2Z_{dc} + 3D_q^2Z_{dc} + 2Z_{L1}};$$
$$G_{vd\_q} = -\frac{3Z_{dc}(D_qV_{dc} - I_qZ_{L1})}{3D_d^2Z_{dc} + 3D_q^2Z_{dc} + 2Z_{L1}};$$

The steady-state operating points in above TFs are obtained from average model and given as

$$V_d = \sqrt{2}\cdot V_{L-P}, (V_{L-P}\text{: line} - \text{to} - \text{phase RMS voltage});$$
$$V_q = 0; I_q = 0; D_d = \frac{3R_{dc}V_d + \sqrt{-24R_{dc}V_{dc}^2R_1 + 9R_{dc}^2V_d^2}}{6R_{dc}V_{dc}};$$
$$I_d = \frac{2V_{dc}}{3R_{dc}D_d}; D_q = -\frac{\omega_L L_1 I_d}{V_{dc}};$$

The dq domain modeling of theL_2 and C_f filter components is derived and integrated with the boost rectifier model to form the complete model of LCL filtered converter, as depicted by its schematic diagram in Fig. 1. For a balanced 3ph star connected system, the admittance matrixes for two elements are modelled as

$$\mathbf{Y}_{L_2\_\mathbf{dq}} = \begin{bmatrix} \frac{Z_{L_2}}{\omega^2L_2^2+Z_{L_2}^2} & \frac{\omega L_2}{\omega^2L_2^2+Z_{L_2}^2} \\ \frac{-\omega L_2}{\omega^2L_2^2+Z_{L_2}^2} & \frac{Z_{L_2}}{\omega^2L_2^2+Z_{L_2}^2} \end{bmatrix}; (Z_{L_2} = L_2s + R_2). \tag{4}$$

$$\mathbf{Y}_{C_f\_\mathbf{dq}} = \begin{bmatrix} \frac{1}{Z_{C_f}} & -\omega C_f \\ \omega C_f & \frac{1}{Z_{C_f}} \end{bmatrix}; Z_{C_f} = \frac{1}{C_fs}. \tag{5}$$

From (1), (2), (4) and (5), the equivalent diagram of power stage in Fig. 1 can be expressed as in Fig. 3, and after simplification, the TF between control to grid-side current ($\hat{i}_{2d}$ and $\hat{i}_{2q}$) can be obtained by forcing the grid voltage input $\mathrm{v}_{g\_\mathbf{dq}}$ to zero

$$\mathbf{G}_{i_2\_d\_\mathbf{dq}} = \frac{\mathbf{G}_{i_1\_d\_\mathbf{dq}}\mathbf{Y}_{L_2\_\mathbf{dq}}}{\mathbf{Y}_{L_2\_\mathbf{dq}} + \mathbf{Y}_{C_f\_\mathbf{dq}} + \mathbf{Y}_{r\_\mathbf{dq}}}. \tag{6}$$

TABLE I
SYSTEM DESIGN PARAMETERS

Inverter Parameters			
Parameter	Value	Parameter	Value
Power rating, P_o	10 kW	Grid voltage, V_{L_L}	380 V
Output voltage, V_{DC}	600 V	Switching freq., f_{sw}	12.5 kHz
Line freq., f_L	50 Hz	Sampling freq., f_s	25 kHz
Load resistance, R_{dc}	37 Ω	DC-link cap., C_{dc}	770 μF
LCL Filter Parameters			
Inverter-side L_1, R_1	1.2 mH; 0.2 ohm		
Filter capacitance, C_f	8 μF		
	Grid-side L_2, R_2		
Case 1: Typical LCL filter	0.906 mH; 0.15 Ω		
Case 2: Integrated LCL filter	4.3 mH; 0.616 Ω		

A 10 kW three-phase VSC system is designed to achieve a unity power factor, with its parameters detailed in Table I. L_1 and C_f were selected to limit the converter-side current ripple and to achieve an approximately unity power factor for the input supply, respectively following guidelines given in [4]. Additionally, the DC-link capacitor was chosen to ensure the DC output voltage ripple remains below 6% of the specified output voltage. The LCL filter design is categorized into two groups based on grid-side inductors: typical and integrated LCL filters. In the typical LCL filter design, L_2 is set independently to maintain the ripple current at 2% and the THD below 5%, as per IEEE Std. 519-2014 [11]. In contrast, the integrated filter's L_2 is determined by the specifications of the IM, as previously discussed, with the values provided in Table I. The equivalent block diagram of the LCL-filtered VSC system, which includes the damping loop, current loop, and outer voltage loop, is depicted in Fig. 4. $G_{del} = e^{-1.5sT_s}$, representing the sampling and computational delay of the control loop, is modeled and simplified using a second-order Padé approximation. The frequency response of the direct component TFs of open loop plant$G_{del}.G_{i_2d_dd}$ for both typical and integrated filter designs is plotted in Fig.5a, revealing pronounced resonance peaks at a lower resonance frequency in the integrated filter design due to the higher grid-side filter inductance. Grid current feedback active damping method is employed for resonance suppersion by measuring the grid-side current and feeds it back through a negated high-pass filter compensator, $G_{HPF}(s)$ [12], and combined with the current controller's output. Regulation of the input grid current is achieved by a Proportional-Integral (PI) current controller, denoted as $\mathbf{G}_{cid\_\mathbf{dq}}$. The converter's output voltage is regulated based on the DC voltage measurements, which are regulated by a PI voltage controller G_{cv}, where V_{dc}^* is the reference DC voltage and $i_{2\_\mathbf{dq}}^*$ is the reference current derived from the voltage controller. It is important to note that reference reactive current is set to 0 for unity power factor. The TFs for the damping compensator is provided as

$$G_{HPF} = -\frac{k_{ad}s}{s+\omega_{ad}} \tag{7}$$

where k_{ad} and ω_{ad} are HPF gain and cut off frequency, respectively. The design parameters of the damping compensators have been chosen to effectively suppress the resonant frequency components in both cases. The controller parameters are graphically determined, and their values are selected to ensure good stability and performance, while maintaining comparable stability characteristics for both filter designs to allow for a fair comparison. The parameters for the designed compensators and controllers are summarized in Table II. Given that the current loop has faster dynamics compared to the slower dynamics of voltage control, which plays a crucial role in overall control loop stability, it is necessary to analyze the frequency response of the system with an open current loop and a closed damping loop. The open

TABLE II
DESIGNED CONTROLLERS AND COMPENSATORS PARAMETERS

	Damping compensator		Current controller		Voltage controller	
	k_{ad}	ω_{ad}	k_{ip}	k_{ii}	k_{vp}	k_{vi}
Case 1	0.04	40212	0.0046	7.15	0.7	6
Case 2	0.163	40212	0.0079	9.81	0.29	8.1

$$G_{id\_dd} = -\frac{3Z_{dc}\omega I_d D_q L_1 + 3Z_{dc}Z_{L_1}I_dD_d + 3Z_{dc}D_q^2V_{dc} + 2Z_{L_1}V_{dc}}{3Z_{dc}Z_{L_1}D_d^2 + 3Z_{dc}Z_{L1}\mathrm{D}_q^2 + 2\omega^2L_1^2 + 2{Z_{L_1}}^2};$$

$$G_{id\_qq} = -\frac{3D_d^2V_{dc}Z_{dc} + 3D_qI_qZ_{dc}Z_{L1} + 2V_{dc}Z_{L_1}}{3Z_{dc}Z_{L_1}D_d^2 + 3Z_{dc}Z_{L_1}D_q^2 + 2\omega^2L_1^2 + 2{Z_{L1}}^2};$$

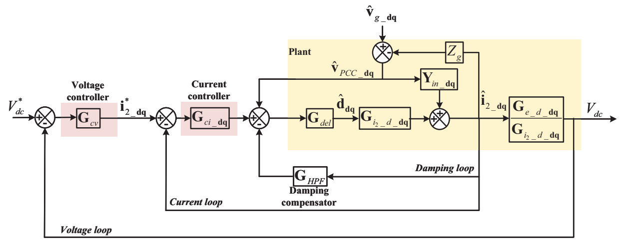
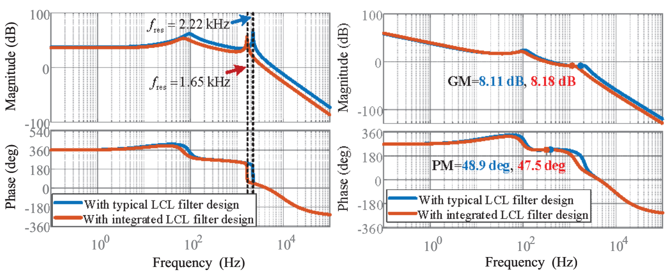


Fig. 4. Equivalent block diagram of LCL-filtered VSC with designed control loops.



(a) (b)

Fig. 5. Frequency response of the system (a) Open loop plant (b) Open current loop system with closed damping loop.

current loop TF of the system with closed inner damping loop is expressed as

$$T_{op}(s) = \frac{G_{ci\_\mathbf{dq}} G_{del} \mathbf{G}_{i_2 d\_\mathbf{dq}}}{1 + G_{HPF} G_{del} \mathbf{G}_{i_2 d\_\mathbf{dq}}} \tag{8}$$

The Bode diagram presented in Fig. 5a shows the frequency response of direct component TF of (8). It can be noticed that the opted values of damping compensator and current controllers suppress the resonance peak and ensure the good stability margins for both cases.

IV. Input Impedance and Stability Analysis

To investigate the input impedance seen by the utility grid and analyse stability characteristics at the PCC terminal, the first step is to calculate the admittance TF matrix of the LCL-filtered VSC without the control structure. This calculation is achieved by simplifying the equivalent block diagram shown in Fig. 4 and setting the control input signal $\hat{\mathbf{d}}_{\mathbf{dq}}$ to 0. The plant admittance TF matrix $\mathbf{Y}_{in\_\mathbf{dq}}$, representing the relationship between the grid current and the PCC voltage, is then calculated as

$$\mathbf{Y}_{in\_\mathbf{dq}} = \frac{\mathbf{Y}_{L_2\_\mathbf{dq}} \left(\mathbf{Y}_{C_f\_\mathbf{dq}} + \mathbf{Y}_{i_1\_\mathbf{dq}} \right)}{\mathbf{Y}_{L_2\_\mathbf{dq}} + \mathbf{Y}_{C_f\_\mathbf{dq}} + \mathbf{Y}_{i_1\_\mathbf{dq}}} \tag{9}$$

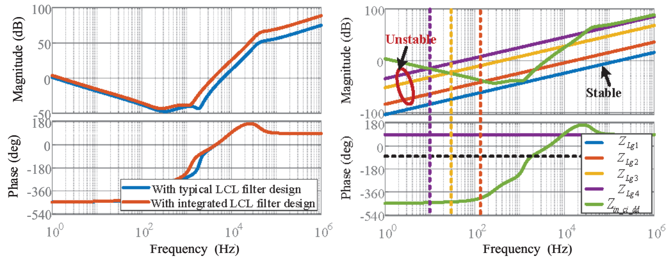
The PCC voltage feedforward is usually employed to improve the dynamic response of the input grid current under transient conditions. Consequently, to evaluate the impedance characteristics of the motor-integrated LCL filter and the stability of the control loop against grid impedance variations, the input impedance is calculated with the overall control structure. The input impedance and stability are first analyzed by closing the current loop and then by closing the outer voltage loop, simplifying Fig. 4. The input admittance TF matrix, considering the closed current and damping loops (assuming $\hat{\mathbf{i}}_{2_{\mathbf{dq}}} = 0$), is derived as

$$\mathbf{Y}_{in\_ci\_\mathbf{dq}} = \frac{\mathbf{Y}_{in\_\mathbf{dq}} + \mathbf{G}_{del}.\mathbf{G}_{i_2\_d\_\mathbf{dq}}}{1 + \left(\mathbf{G}_{ci\_\mathbf{dq}} - G_{HPF} \right) .\mathbf{G}_{del}.\mathbf{G}_{i_2\_d\_\mathbf{dq}}} \tag{10}$$

Similarly, the admittance TF after closing the voltage loop (assuming $V_{dc}^* = 0$) can be calculated and is detailed on the next page. The Bode diagram in Fig. 6a displays the frequency response of the direct component impedance function (i.e., $Z_{in\_ci\_dd} = Y_{in\_ci\_dd}^{-1}$) with the current loop closed for both filter cases. The impedance is the same in the low-frequency region for both LCL filter cases, with a higher impedance from medium to high frequency ranges for the integrated LCL filter, indicating better suppression capability for high-frequency harmonics. However, a phase shift in the phase curve exceeding $-180°$ degrees for both scenarios results in approximately similar stability characteristics. This is illustrated by the Bode diagram in Fig. 6b, plotted for $Z_{in\_ci\_dd}$ for the integrated LCL filter, considering different grid impedances with four different values of grid inductances $L_g = \{1\mu\text{H}, 10\mu\text{H}, 400\mu\text{H}, 3\text{mH}\}$. The control loop becomes unstable for $Z_g > Z_{g1}$ due to the phase curve shifting beyond $-180°$ at the magnitude intersection between different Z_g and $Z_{in_c i_d d}$. Further, the impedance stability is investigated by applying the generalized Nyquist criterion (GNC) on the impedance ratio $Z_{ratio\_ci\_i} = Z_{gi}.Y_{in_c i_d d}$, as shown in Fig. 7a. All impedance ratios encircle the -1 critical point, indicating the instability of the control except for Z_{g1}, which does not encircle the -1 critical point, suggesting stability as displayed in the zoomed-in view of Fig. 7b.

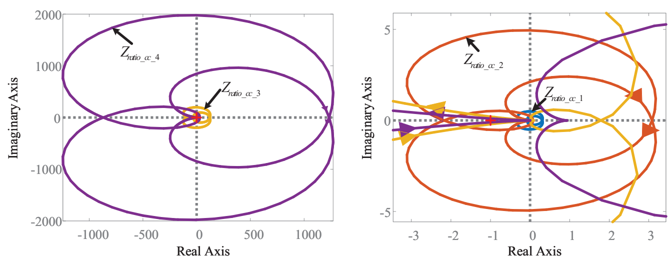
The Bode diagram in Fig. 8a shows the input impedance (i.e., $Z_{in\_cv\_dd} = Y_{in\_cv\_dd}^{-1}$) response with the voltage loop closed. The impedance magnitude trend is similar to that observed with the current loop closed, showing similar magnitudes in the low-frequency region and higher impedance magnitudes in the medium and high-frequency regions. This indicates comparatively low medium to high-frequency oscillations in grid current with the integrated LCL filter case. Importantly, the phase curve shift is less than $-180°$ in the phase plot,

$$\mathbf{Y}_{in\_cv\_\mathbf{dq}} = \frac{\mathbf{Y}_{in\_\mathbf{dq}} + \mathbf{G}_{del}.\mathbf{G}_{i_2\_d\_\mathbf{dq}}}{1 + [\mathbf{G}_{del}.G_{HPF}.\mathbf{G}_{i_2\_d\_\mathbf{dq}} + \mathbf{G}_{ci\_\mathbf{dq}}.\mathbf{G}_{del}\left(\mathbf{G}_{i_2\_d\_\mathbf{dq}} + G_{cv}.\mathbf{G}_{e\_d\_\mathbf{dq}}\right)]} \tag{11}$$



(a) (b)

Fig. 6. Closed current loop impedance frequency response for (a) Input impedance comparison (b) Grid impedances vs Input impedance response for integrated LCL-filter.

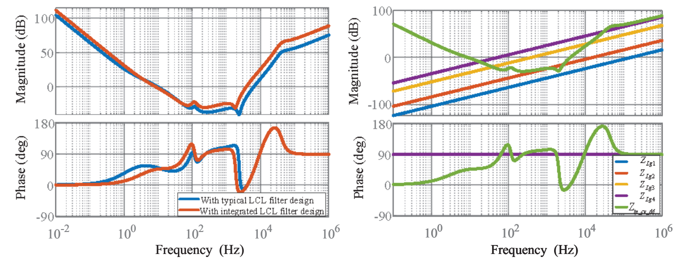


(a) (b)

Fig. 7. Closed current loop impedance ratio plot (a) Input impedance vs different grid impedances (b) Zoomed-in view.

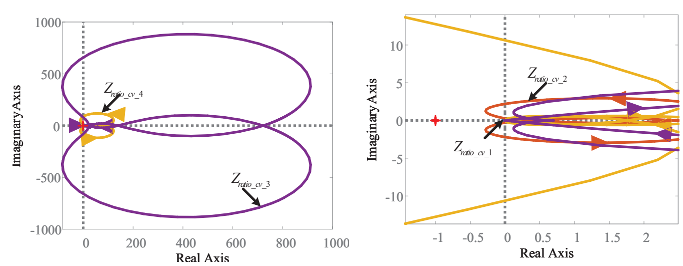
indicating that the control loop will have no instability risk under impedance variation. The Bode diagram in Fig. 8b depicts the frequency response of $Z_{in\_cv\_dd}$ and different Z_{gi}. It can be observed that there are no -180° phase crossings during the magnitude intersection of Z_{gi} and $Z_{in\_cv\_dd}$, ensuring overall control stability for different values of Z_g. This stability has been further verified by applying the Generalized Nyquist Criterion (GNC) to the impedance ratio $Z_{ratio\_cv\_i} = Z_{gi}.Y_{in\_cv\_dd}$ with varying Z_g, as shown in the Nyquist diagram in Fig. 9. Clearly, from Fig. 9a and Fig. 9b, none of the impedance ratio functions encircle the critical point -1 for different values of Z_g, and the control loop will remain stable.

In summary, the integrated LCL filter provides better oscillation rejection in the medium to high-frequency range compared to the typical LCL filter design, and the control loop remains stable for both cases with the voltage loop closed under Z_g variations. This contrasts with the potential instability under Z_g variations when only the current loop with an inner damping loop is considered.



(a) (b)

Fig. 8. Closed voltage loop impedance frequency response (a) Input impedance comparison (b) Grid impedances vs Input impedance response for integrated LCL-filter.



(a) (b)

Fig. 9. Closed voltage loop impedance ratio plot (a) Input impedance vs different grid impedances (b) Zoomed-in view.

V. Experimental Validation

The integrated LCL-filtered OBC and its controllers are validated using a Typhoon hardware-in-loop (HIL) based real-time controller setup, shown in Fig. 10. This system includes the Typhoon HIL device 402, a control board, and an oscilloscope, all operated on a computer system that hosts the OBC model and control systems. The control loops and damping compensators were digitally implemented via bilinear discretization. The model executed in real-time on the Typhoon HIL 402, interfacing with the control board to produce physical responses, while the oscilloscope displayed the resulting waveforms, providing a robust platform for VSC testing and implementation.

Fig. 11 displays steady-state test waveforms for input and output current and voltage. The sinusoidal grid

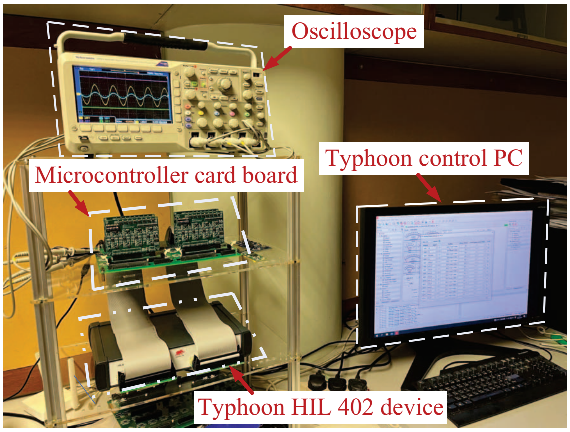


Fig. 10. Hardware-in-loop test set-up.

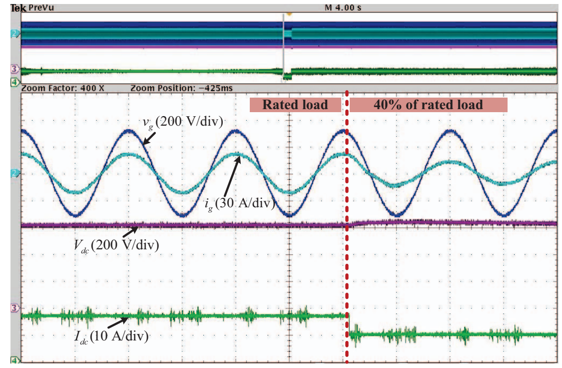


Fig. 12. Test result waveforms under load step change.

current, which is in phase with the grid voltage, ensures a unity power factor and high current quality, evidenced by a 1.43% THD at the rated output DC voltage and current. Additionally, the resonance frequency component is eliminated in the grid current, demonstrating the effectiveness of the designed control structure for the VSC. Furthermore, the control design is tested under a load step change, as depicted in Fig. 12. It is observed that when the load is reduced from rated load to 40% of the rated load, the reference output DC voltage is effectively maintained without any transient oscillations in the grid current and voltage waveforms.

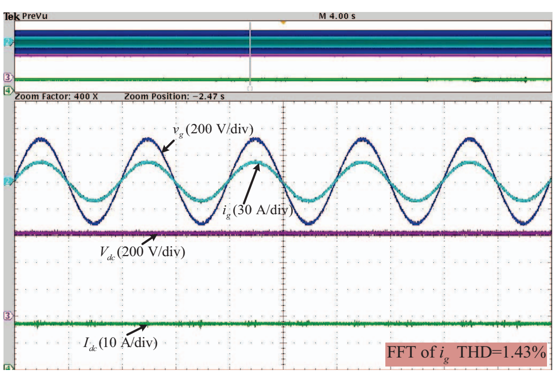
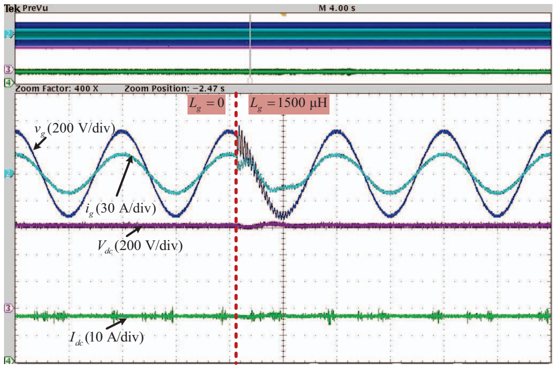


Fig. 11. Test result waveforms during steady-state operation.

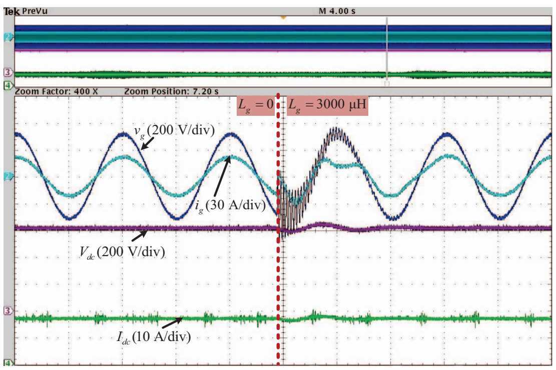
To verify the input impedance stability analysis, the control loop performance is tested under grid impedance variations. The control loop remains stable, consistently injecting sinusoidal grid current and regulating the output DC voltage at the reference value, even when the grid inductance is switched from 0 to 1500 μH, as depicted in Fig. 13a. Similar stability observations are noted in Fig. 13b where the grid impedance transitions from 0 to 3 mH, aligning with analytical stability findings for the integrated LCL-filtered OBC. However, strong transient oscillations during the transition instant are clearly visible in Fig. 13, and these oscillations are get amplified at higher grid inductances as shown in Fig. 13b. This behavior is expected due to the significant negative impedance region in the input impedance function of the VSC with all designed control loops. These test results validate the theoretical findings. It is important to note that although the designed control remains stable with grid impedance variations, significant oscillations during grid impedance changes may trigger instability. Therefore, a suitable compensation strategy is necessary to mitigate the stability risk associated with the negative impedance region.

VI. Conclusion

An integrated LCL-filtered VSC for an OBC in EVs has been developed, utilizing the traction motor's stator winding as an input filter for the front-end rectifier stage during charging mode. The control structure, including damping, current, and outer voltage loops, has been designed based on the developed small-signal dq frame model. The input impedance characteristics of the VSC were investigated with both current and voltage loops closed and compared with a typical LCL filter design. The integrated LCL filter design exhibits higher impedance at medium and high-frequency regions, resulting in relatively fewer oscillations during transients. The input impedance investigation demonstrates that the control structure for both filter designs remains stable under grid impedance variations, despite a negative impedance frequency region which induces transient oscillations, underscoring the need for a suitable compensation strategy. These findings have been validated through real-time control hardware-in-the-Loop testing.



(a)



(b)

Fig. 13. Test result waveforms at (a) Grid inductance change 0 to 1500 μH (b) Grid inductance change 0 to 3 mH.

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